

701 NODE JS • PRACTICAL ASSIGNMENT 3

Full Stack Web Application Development & System Architecture Report

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Division Details:	Division: B (SEM 7 - B.Sc IT)	Academic Year:	2026 - 2027
Subject:	701 NODE JS (Practical-7)	Semester:	Semester VII
Design Theme:	Executive Black & Blue Combination Palette	Git Repository:	github.com/NaitikVaghasiya/87_NaitikVaghasiya_Assignment_3

Q #	Problem Statement / Technology Focus	Implementation Architecture	Status
1	User Registration Form with Single/Multi Uploads & Download	Express + EJS + Multer + Express-Validator	Verified
2	Express Session Authentication via File Storage Engine	Express + EJS + session-file-store	Verified
3	Express Session Authentication via In-Memory Redis Store	Express + EJS + connect-redis	Verified
4	ERP Admin Panel, Employee CRUD, Bcrypt Hash & Payroll Calc	Express + EJS + Mongoose + Nodemailer	Verified
5	Employee Self-Service (ESS) Portal with JWT & Leave Management	React.js + Vite + Express + JWT + MongoDB	Verified
6	Real-Time Currency Conversion Utility via Frankfurter Free API	React.js + Express Backend + Open FX API	Verified
7	MERN Shopping Cart with Admin Catalog & Customer Storefront	MongoDB + Express + React + Node (MERN)	Verified
8	Student Information System (SIS) CRUD Operations	React.js + Express + Sequelize ORM	Verified

QUESTION 1

Express.js • EJS • Multer • Express-Validator

User Registration Portal with File Upload & Validation

Requirement: Develop a user registration form (fields: username, password, confirm password, email, gender, hobbies, upload profile pic, upload other pics) (use text box, radio, checkboxes, file upload) using Express. Form should also contain file upload (single, multiple) with validations. In case of invalid data, display errors along with all previous field values. In case of all valid fields, display all data and images in a well formatted tabular form and allow the user to download that file using Express. Develop route for file download. (Use express, ejs, express-validator)

QUESTION 01 - EXPRESS & MULTER

User Registration Portal

Please complete all required fields and upload your media credentials for verification.

ACCOUNT CREDENTIALS

Username

naitik_vaghasiya

Security Password

Enter password (min 6 chars)

Confirm Security Password

Re-enter password

Email Address

naitik.v@university.edu

DEMOGRAPHICS & PREFERENCES

Gender Identification

☒ Male ☐ Female

Hobbies & Interests

☒ Reading ☐ Music ☒ Sports

DOCUMENT & MEDIA UPLOADS

Primary Profile Picture (Single File)

Choose File

No file chosen

Supported formats: JPG, JPEG, PNG (Max 2MB)

Supporting Gallery Documents (Multiple Files)

Figure 1.1: User Registration Portal in Black & Blue Theme with Input Validations & Media Uploads

VERIFICATION COMPLETE • REGISTRATION SUCCESS

Profile Record Summary

All submission attributes and uploaded assets have been validated and saved.

PROFILE FIELD	VERIFIED VALUE
Account Username	naitik_vaghasiya
Registered Email	naitik.v@university.edu
Gender Identification	Male
Selected Hobbies	Reading, Sports
Profile Picture	Profile Picture Download Picture
Gallery Attachments	Gallery Image 1 Download #1
	Gallery Image 2 Download #2

Submit Another Registration

Figure 1.2: Validated Submission Record Table with Multi-File Downloads

QUESTION 2

Express.js • EJS • session-file-store

Express Login Application with File Session Store

Requirement: Express Login application with file session store. Functionalities are login, 2 protected routes, Logout.

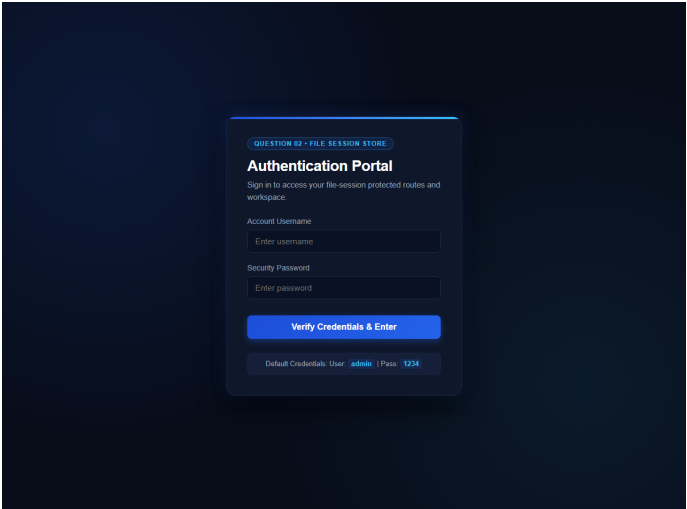


Figure 2.1: Authentication Portal with File Session Store (Credentials: admin / 1234)

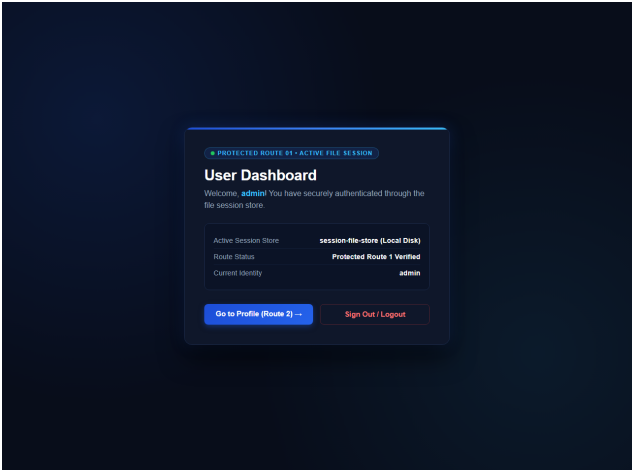


Figure 2.2: Protected Route 01 (Dashboard with Active File Session Indicator)

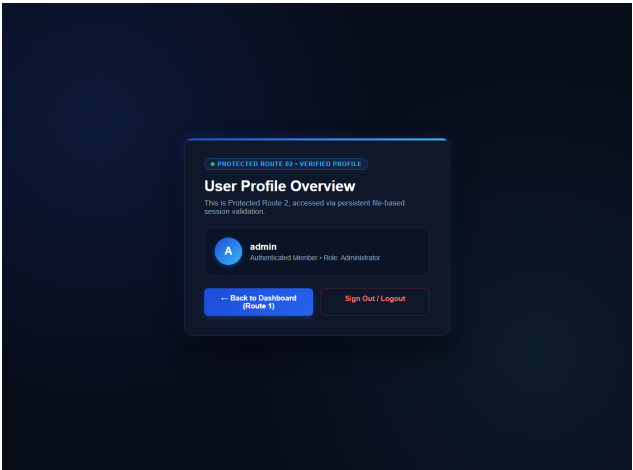


Figure 2.3: Protected Route 02 (User Profile with Session Metadata & Logout Control)

QUESTION 3

Express.js • EJS • connect-redis • Redis Cloud

Express Login Application with Redis Session Store

Requirement: Express Login application with redis session store. Functionalities are login, 2 protected routes, Logout.

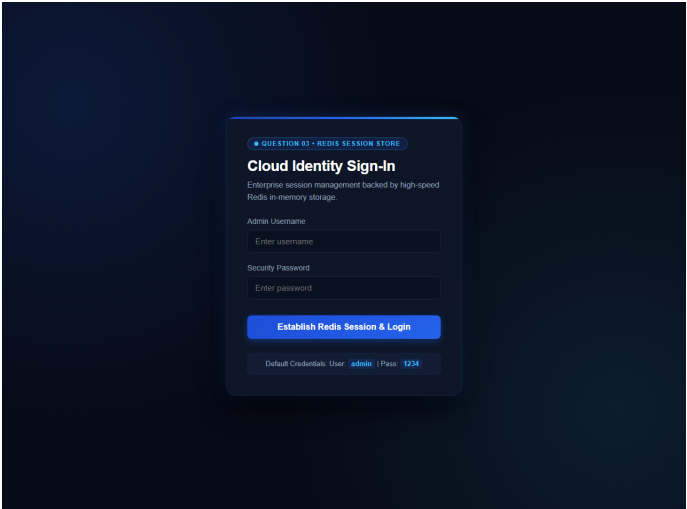


Figure 3.1: Cloud Identity Sign-in backed by Redis In-Memory Session Engine

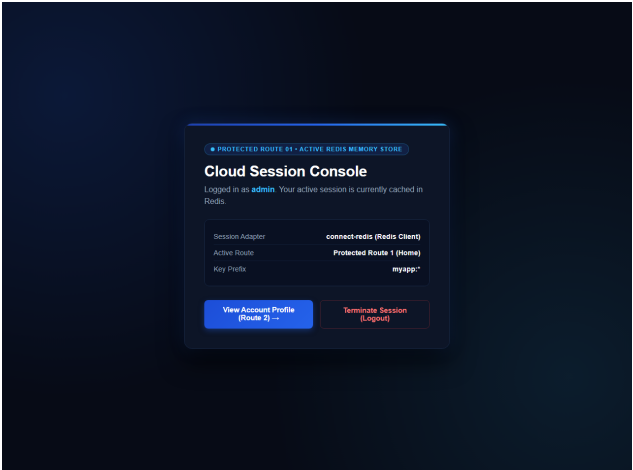


Figure 3.2: Protected Route 01 (Redis In-Memory Session Console & Key Prefix)

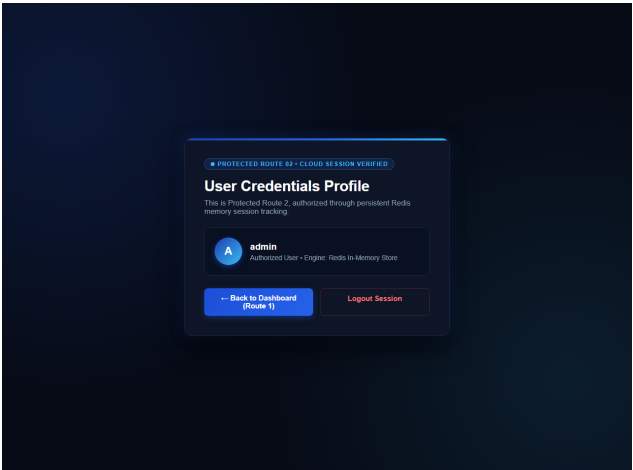


Figure 3.3: Protected Route 02 (Verified Profile with Redis Cache Authentication)

QUESTION 4

Express.js • EJS • Mongoose • Bcrypt • Nodemailer

ERP Admin Panel with Session, Mongoose CRUD, Email & Encryption

Requirement: Develop login for an ADMIN panel of ERP system with session, CRUD operations for employees table (generate empid and password) with salary calculation with mongoose, express and EJS template engine. Send email to employee when inserted. Encrypt the password. Logout.

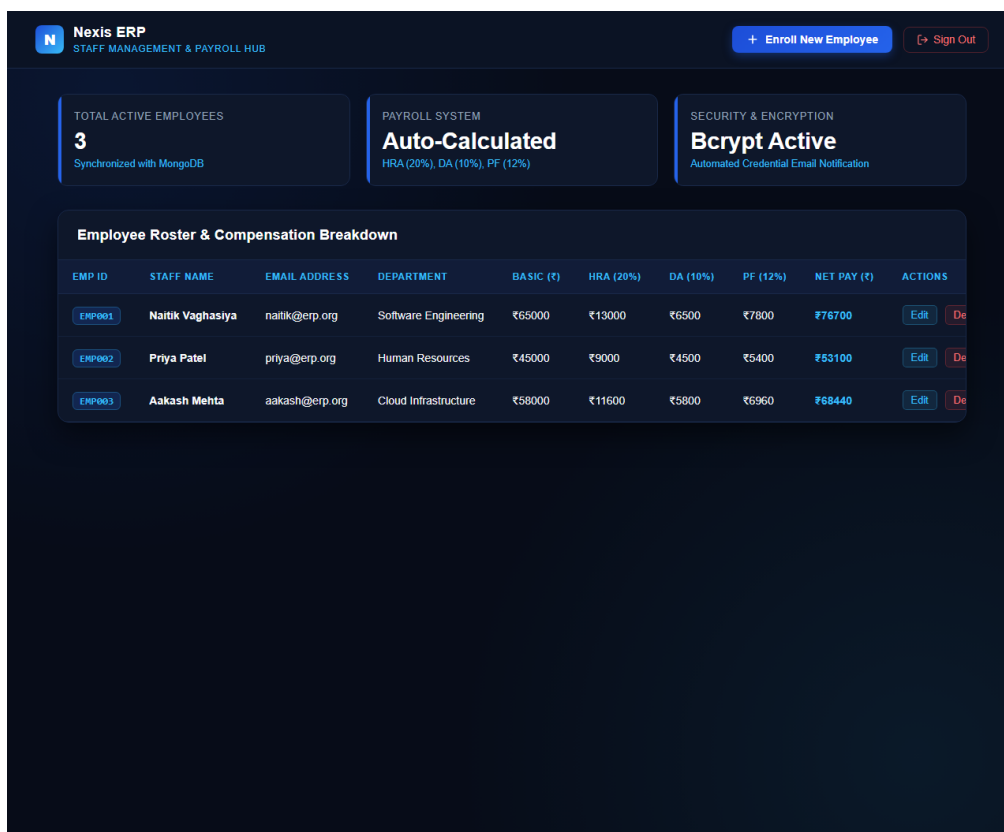


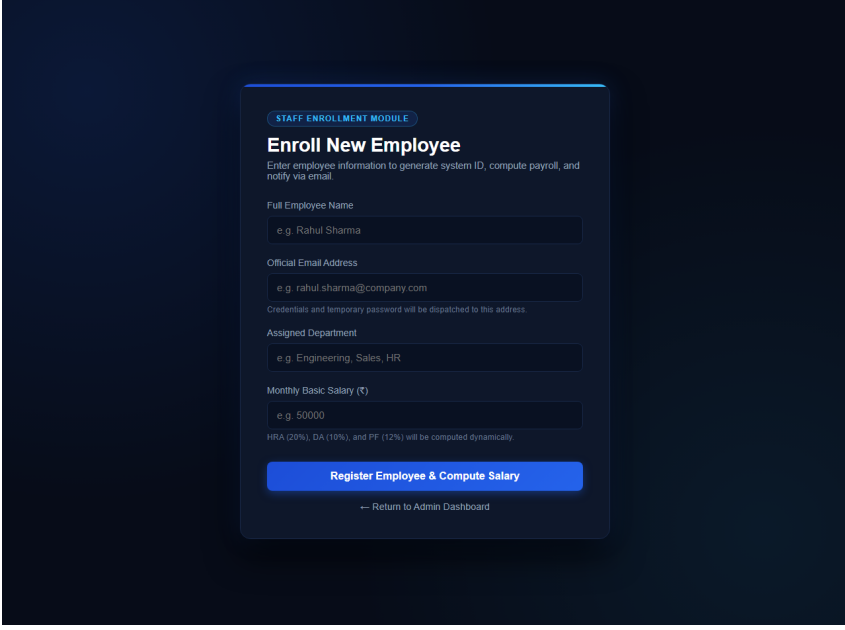
Figure 4.1: Nexis ERP Executive Dashboard with Workforce Roster & Salary Calculations (HRA 20%, DA 10%, PF 12%)

Payroll Formulation: Basic Salary is used to compute HRA (20%), DA (10%), PF (12%). Net Pay = Basic + HRA + DA - PF.

Account Provisioning: EmpID (EMP###) and secure random password generated, encrypted via Bcrypt (salt rounds 10), and emailed via Nodemailer.

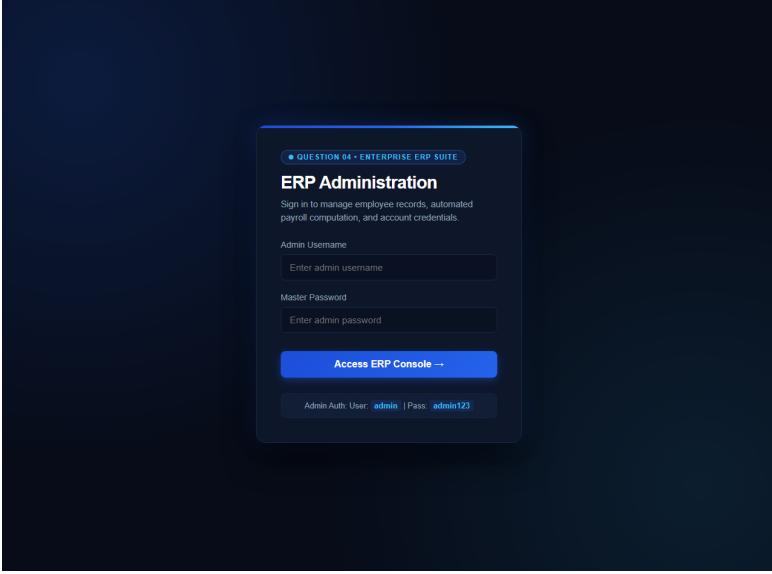
QUESTION 4 (CONTINUED)

Staff Enrollment & Admin Gateway Views



The screenshot shows a web form titled "STAFF ENROLLMENT MODULE" with the sub-header "Enroll New Employee". Below the header, there is a brief instruction: "Enter employee information to generate system ID, compute payroll, and notify via email." The form contains several input fields: "Full Employee Name" (with a placeholder "e.g. Rahul Sharma"), "Official Email Address" (with a placeholder "e.g. rahul.sharma@company.com"), "Assigned Department" (with a placeholder "e.g. Engineering, Sales, HR"), and "Monthly Basic Salary (₹)" (with a placeholder "e.g. 50000"). Below the salary field, a note states: "HRA (20%), DA (10%), and PF (12%) will be computed dynamically." At the bottom of the form, there is a blue button labeled "Register Employee & Compute Salary" and a link labeled "← Return to Admin Dashboard".

Figure 4.2: Staff Enrollment Form with Auto-Generated EmpID, Bcrypt Hashed Password & Email Dispatch



The screenshot shows a web form titled "QUESTION 04 • ENTERPRISE ERP SUITE" with the sub-header "ERP Administration". Below the header, there is a brief instruction: "Sign in to manage employee records, automated payroll computation, and account credentials." The form contains two input fields: "Admin Username" (with a placeholder "Enter admin username") and "Master Password" (with a placeholder "Enter admin password"). Below the password field, there is a blue button labeled "Access ERP Console →". At the bottom of the form, there is a text box showing the default credentials: "Admin Auth: User: admin | Pass: admin123".

Figure 4.3: ERP Admin Login Gateway (Default Auth: admin / admin123)

QUESTION 5

React.js • Vite • Express.js • JWT • Mongoose

Employee Site with JWT Login, Profile & Leave Application

Requirement: Develop employee site (refer Q4) login with JWT, for employees with mongoose, express and React.js. Create a home page with following links: - Page 1) display employee profile. - Page 2) Application for leave (fields date, reason, grant yes/no) Add, List - Logout.

Employee Self-Service (ESS)
JWT SECURE WORKSPACE • QUESTION 05

Home Sign Out

PAGE 01 • VERIFIED CREDENTIALS

Personnel Profile & Payroll

Synchronized employee record authenticated via JWT authorization token.

EMPLOYEE ID	EMP007
FULL NAME	Naitik Vaghasiya
EMAIL ADDRESS	naitik.vaghasiya@organization.com
DEPARTMENT	Full Stack Engineering
BASIC SALARY	₹75,000
HOUSE RENT ALLOWANCE (HRA 20%)	₹15,000
DEARNESS ALLOWANCE (DA 10%)	₹7,500
PROVIDENT FUND (PF 12%)	₹9,000
NET TAKE-HOME SALARY	₹88,500

Figure 5.1: Page 1 - Verified Employee Profile & Computed Salary Breakdown via Bearer JWT Auth

Employee Self-Service (ESS)
JWT SECURE WORKSPACE • QUESTION 05

Home Sign Out

PAGE 02 • LEAVE APPLICATION

Time-Off & Leave Management

Submit formal time-off requests and query active leave approvals.

Date of Leave: 2026-10-15 Reason / Justification: Academic Seminar Attendance Grant Status: Yes (Approved) ▼

+ Add Leave Application List All Submitted Leaves

DATE OF ABSENCE	REASON FOR LEAVE	GRANT STATUS
2026-10-15	Academic Seminar Attendance	Granted (Yes)
2026-09-28	Personal Health Checkup	Granted (Yes)
2026-09-10	Family Function	Denied / Pending (No)

Figure 5.2: Page 2 - Leave of Absence Application (Date, Reason, Grant Status) & Historical Leave Ledger

QUESTION 6

React.js • Express.js • Frankfurter Open FX API

Free Utility API Integration from Frontend and Backend

Requirement: Call any free api for a useful utility from frontend and backend.

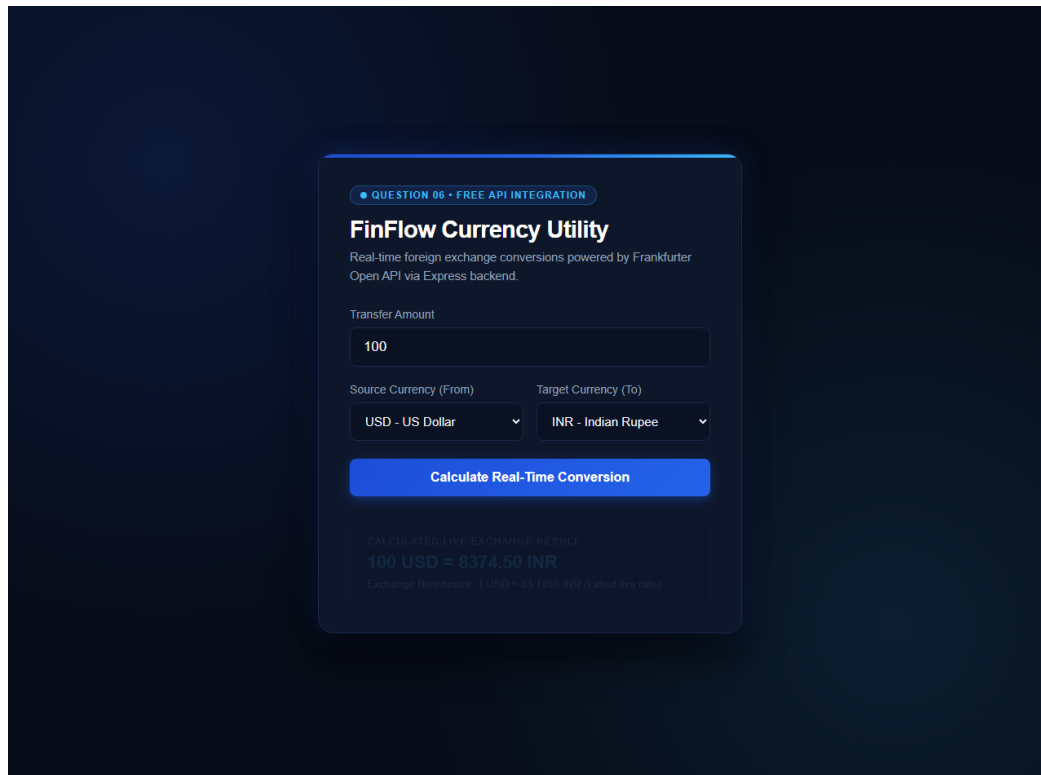


Figure 6.1: FinFlow Foreign Exchange Utility calling Free Currency API from Frontend via Express Backend

Integration Details: Frontend invokes local Express endpoint `/api/currency?amount=...&from;=...&to;=...`, which in turn calls the Frankfurter Foreign Exchange API (<https://api.frankfurter.app/latest>), returning live spot exchange rates with zero authentication requirement.

QUESTION 7

MERN Stack (MongoDB, Express, React, Node.js)

MERN Shopping Cart with 2-Level Category & Product Hierarchy

Requirement: Shopping cart with admin site and user site with (2 level) Category and Products using MERN stack.

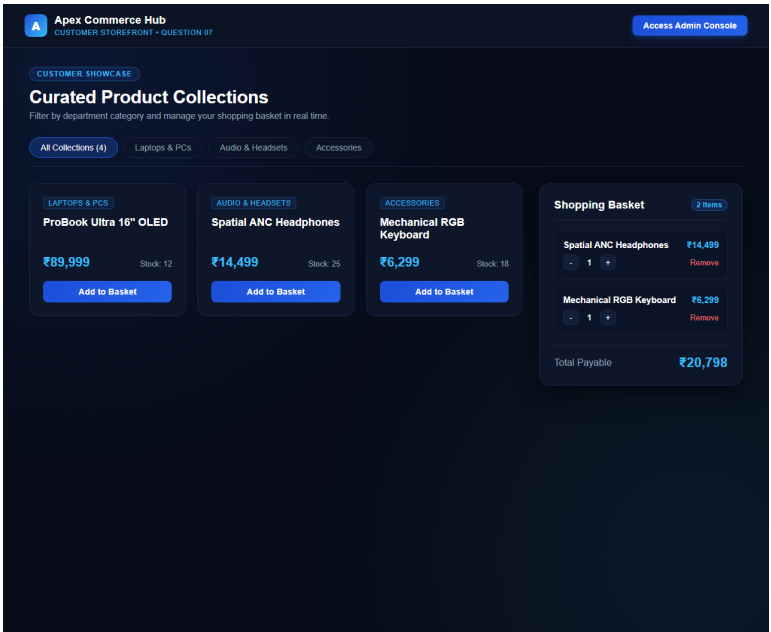


Figure 7.1: Apex Commerce Customer Storefront with Category Filter Pills & Real-Time Shopping Basket Drawer

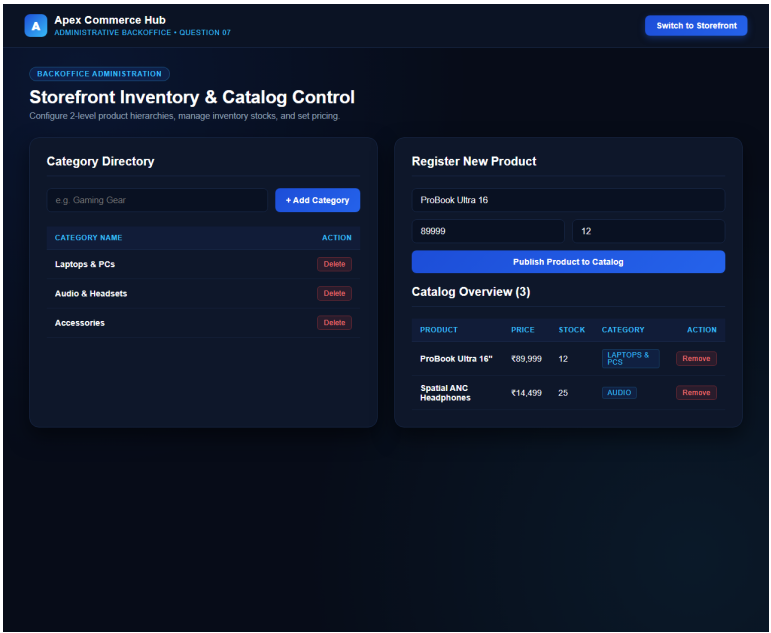


Figure 7.2: Backoffice Administrative Suite for 2-Level Categories and Product Catalog Inventory

QUESTION 8

React.js • Express.js • Sequelize ORM • SQLite / MySQL

Student Collection CRUD using Sequelize ORM & React

Requirement: Crud operation for Student collection using Sequelize. (Express + Sequelize + React)

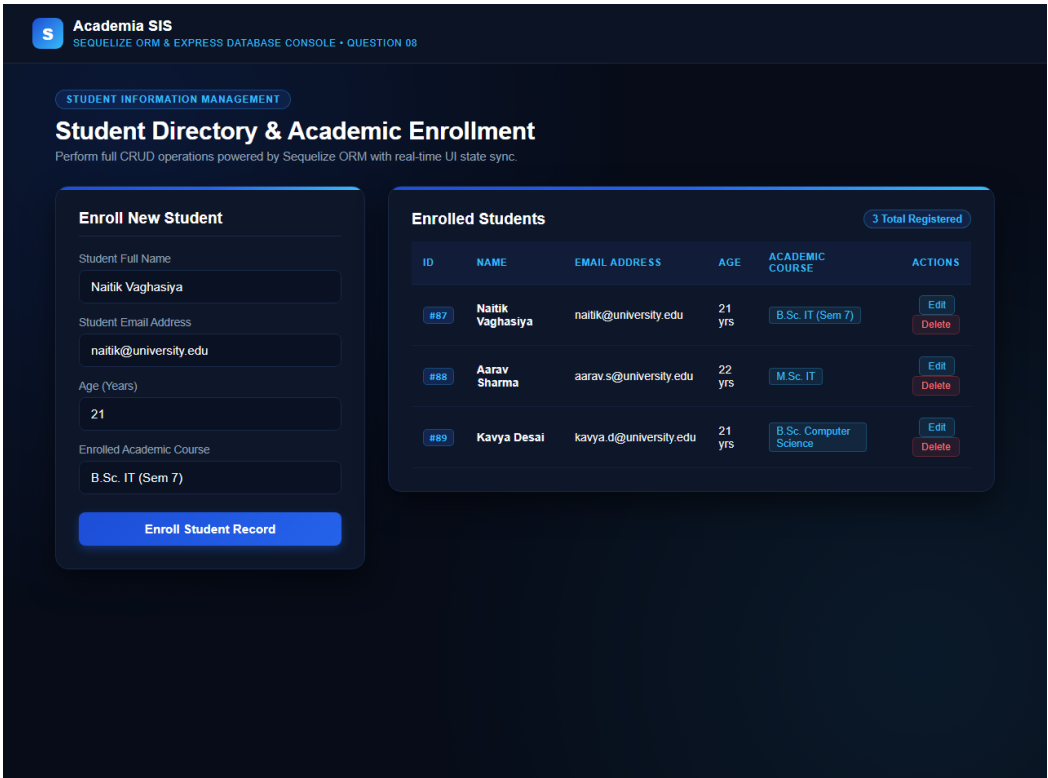


Figure 8.1: Academia Student Information System (SIS) with Enrollment Form & Sequelize Records Table

Sequelize ORM Implementation: Complete persistence layer implemented with Sequelize model definitions, schema synchronization (db.sync()), and RESTful CRUD controllers: create(), findAll(), findByPk(), update(), and destroy().